

1. Extend $\deg(0) = -\infty$.

$$\begin{aligned} f &= qg + r \\ f(t) &= f_d t^d + \cdots, f_d \neq 0 \\ q(t) &= a_n t^n + \cdots, a_n \neq 0 \\ g(t) &= b_m t^m + \cdots, b_m \neq 0 \end{aligned}$$

For existence:

$$\begin{aligned} P(n): \text{ for every } f \in F[t] \text{ with } \deg(f) = n, n \in \mathbb{N}, g \neq 0, \exists q, r: f &= qg + r, \deg(r) < \deg(g) \\ f = 0: 0 &= 0 \cdot g + 0 \\ \deg(0) &= -\infty < \deg(g) \end{aligned}$$

- base

$$\begin{aligned} P(n) \text{ is true for } n < \deg(g) \\ q = 0, r = f, f &= qg + r, \deg(r) = \deg(f) = n < \deg(g) \end{aligned}$$

- induction step, assume for every $n \leq N$: $P(n)$ is true, prove it for $N + 1$.

$$\begin{aligned} g_m \in F, F \text{ is a field, } g_m \neq 0 &\implies \exists g_m^{-1} \\ \text{let } a &= g_m^{-1} f_{N+1} \\ \text{let } f' = f - ag_m t^{N+1-m} &\implies \deg(f') < \deg(f) \implies f' = q'g + r \implies f = (q' + at^{N+1-m})g + r \end{aligned}$$

which concludes the induction.

For uniqueness assume that $\exists q', r': q \neq q'$ or $r \neq r', f = q'g + r'$.

$$\begin{aligned} f &= qg + r, f = q'g + r' \\ qg + r - (q'g + r') &= 0 \\ (q - q')g + (r - r') &= 0 \\ \deg(r) < \deg(g), \deg(r') < \deg(g) &\implies \deg(r - r') < \deg(g) \implies \deg(r' - r) < \deg(g) \end{aligned}$$

Consider two cases:

$$q = q' \implies r = r'$$

contradiction with initial assumption

$$\begin{aligned} q \neq q' &\implies \deg(q - q') \geq 0 \implies \deg((q - q')g) \geq \deg(g) \\ (q - q')g &= r' - r \implies \deg((q - q')g) = \deg(r' - r) \end{aligned}$$

which is a contradiction since $LHS \geq \deg(g), RHS < \deg(g)$, therefore $q' = q, r' = r$.

2. Let's use Euclidean algorithm for finding $\gcd(g, f)$:

- (a) if $f = 0$ result is g
- (b) otherwise result is $\gcd(r, g)$ where $f = qg + r$.

Such q, r exist (see problem 1), $\deg(r) < \deg(g)$. Therefore steps of Euclidean algorithm will look like:

$$\begin{aligned} f &= q_0g + r_0 \\ g &= q_1r_0 + r_1 \\ r_0 &= q_2r_1 + r_2 \\ &\dots \\ r_{k-1} &= q_{k+1}r_k + r_{k+1}, r_{k+1} = 0 \end{aligned}$$

Since $\deg(r_k)$ decreases with each step this algorithm is going to terminate in a result d .

Let's prove that $d \mid f, d \mid g$. To do that we will do reverse Euclidian algo steps to reconstruct f, g from d using existing coefficients q_i, r_i :

$$\begin{aligned} r_k &= d, r_{k+1} = 0 \\ d \mid r_i, d \mid r_{i+1}, r_{i-1} = q_{i+1}r_i + r_{i+1} &\implies d \mid r_{i-1} \\ \text{therefore } d \mid r_1, d \mid r_0 &\implies d \mid g, d \mid f \end{aligned}$$

Now let's consider an arbitrary c , such that $c \mid f, c \mid g$.

$$\begin{aligned} c \mid f, c \mid g &\implies c \mid r_0 \\ c \mid g, c \mid r_0 &\implies c \mid r_1 \\ &\dots \\ c \mid r_k, c \mid r_{k+1} &\implies c \mid r_{k+2} \end{aligned}$$

Since $\exists k: d = r_k$ then $c \mid d$ for arbitrary c . Therefore since $d \mid f, d \mid g, d = \gcd(f, g)$.

Now since d must be monic let's consider $d(t) = d_0 + d_1t + \dots + d_nt^n$. Since $d_n \in F, F$ — field $\implies \exists d_n^{-1} \implies d' = d_n^{-1}d$ is monic.

3. a) Assume $\gcd(\pi, f) = c$, then $c \mid \pi, c \mid f$. Since π is irreducible then $\pi = cd$, and either c or d is a unit. There are two possibilities:
- c is a unit. But then only monic unit is 1 and $\gcd(\pi, f) = 1$.
 - d is a unit, c is an irreducible polynomial. But since π is irreducible then c and π are from the same equivalence class. But then $\gcd(\pi, f) = \pi$ and $\pi \mid f$ which is a contradiction.
- b) $\pi \nmid fg$. Assume $\pi \nmid f$ and $\pi \nmid g \implies \gcd(\pi, f) = 1$.

$$\begin{aligned} 1 &= \pi x + fy \\ g &= \pi gx + fgy \\ \pi \mid \pi gx, \pi \mid fgy &\implies \pi \mid g \text{ — contradiction} \end{aligned}$$

therefore $\pi \mid f$ or $\pi \mid g$.

c) define the following algorithm. Let's say $fact(n)$ is a finite product of irreducible polynomials:

- if f is irreducible $fact(f) = f$
- if f is reducible $\exists h, g: f = hg \implies fact(f) = fact(h)fact(g)$.
- $\deg(h) < \deg(f), \deg(g) < \deg(f), \deg \in \mathbb{N}$ and $<$ on $\mathbb{N}$ is well-founded. $NSTEPS(call)$ — number of steps needed for termination.
 $P(n) = \forall f: \deg(f) = n, NSTEPS(fact(f)) < \infty$
 $P(1)$ is true since such f are irreducible.
Assume $P(n)$ is true for all $n \leq N$. Prove for $N + 1$.
For arbitrary $f: \deg(f) = N + 1, NSTEPS(fact(f)) = 1$ if it's irreducible or $NSTEPS(fact(f)) = NSTEPS(fact(g)) + NSTEPS(fact(h)) + 1$ otherwise $\implies P(N + 1) \text{ true}$.

d) consider arbitrary π_i . If $\pi_i \mid f$ then from 3.b there are two possibilities:

- $\exists \rho_j: \pi_i \mid \rho_j$. This means that ρ_j is either reducible which is a contradiction or $\rho_j = h\pi_i$ where $h \in F[t]^\times$, which means π_i and ρ_j are associates.
- $\pi_i \mid v$, but this is impossible because $\deg(v) < \deg(\pi_i)$.

To prove 1-to-1 matching of π_i and ρ_j and $r = s$ do the following:

- Remove u from the left hand side by multiplying both sides by u^{-1} and reassigning $v = vu^{-1}$. Since all π_i and ρ_j are monic then their products leading coefficients are equal to 1 which means $v = 1$ as well and can be removed.
- Suppose $\pi_1 \cdots \pi_r = \rho_1 \cdots \rho_s$. Then $\exists j: \pi_1 = \rho_j$.
 $F[t]$ is an integral domain $\implies$ multiplication is commutative $\implies$ we can reorder and get

$$\pi_1 \cdots \pi_r = \rho_j \rho_1 \cdots \rho_{j-1} \rho_{j+1} \cdots \rho_s$$

$F[t]$ is an integral domain $\implies$ 0 has no non-zero divisors which allows cancellation and reindexing:

$$\pi_2 \cdots \pi_r = \rho_2 \cdots \rho_s$$

Since number of factors is finite this process repeats until no more factors left on left or right side of the equality.

- Assume $r \neq s$, for example $r < s$ then after this process ends we get

$$1 = \rho_k \cdots \rho_s$$

Which is impossible because $\deg(1) = 0, \deg(\rho_k \cdots \rho_s) \geq 1$.

Similarly for $r > s$. Contradiction.

4. • $F[t]/(m)$ is a field, assume m is reducible. Then $m = ab, 1 \leq \deg(a), \deg(b) < \deg(m), [m] = [ab] = [a][b] = 0, [a] \neq 0, [b] \neq 0$, but that means that $[a], [b]$ are non-zero zero divisors, contradiction.
 • m is irreducible. Consider arbitrary $[f] \neq [0]$.

$$[f] \neq [0] \implies m \nmid f \implies \gcd(m, f) = 1 \implies \exists x, y \in F[t]: 1 = mx + fy \implies [fy] = [1] \implies [f][y] = [1] \implies [y] = [f]^{-1}$$

Therefore for arbitrary non-zero $[f]$ exists inverse.

5. a) For arbitrary element $[f] \in K$ euclidian division for representative gives $f = qm + r$. Assume:

$$f = q_1m + r_1 = q_2m + r_2, q_1 \neq q_2, r_1 \neq r_2$$

then

$$\begin{aligned} q_1m + r_1 - (q_2m + r_2) &= 0 \\ m(q_1 - q_2) + (r_1 - r_2) &= 0 \\ \deg(m) > \deg(r_1 - r_2), \deg(m) \geq 1 &\implies q_1 = q_2, r_1 = r_2 \end{aligned}$$

- b) Split $\mathbb{F}_p[t]$ into two disjoint sets:

$$\begin{aligned} S_1 &= \{f: \deg(f) < d\}, S_2 = \{f: \deg(f) \geq d\} \\ \text{by axiom of replacement } T_1 &= \{[f]: f \in S_1\}, T_2 = \{[f]: f \in S_2\} \end{aligned}$$

From 5.a $T_2 \subseteq T_1$. For arbitrary $[f] \in T_1$ its representative has form $f(t) = a_0 + a_1t + \cdots + a_{d-1}t^{d-1}$ therefore $|T_1| = \prod_{i=0}^{d-1} |\mathbb{F}_p|$ since every coefficient a_i is independent and can take any value from $\mathbb{F}_p$.
 $T_1 \cup T_2 = K, T_1 \cup T_2 = T_1, |T_1| = p^d \implies |K| = p^d$.

c)

$$\begin{aligned}
[t]^d &= [t] \times \cdots \times [t] (\text{d times}) = [t^d] \implies m(\alpha) = [t^d] + c_{d-1}[t^{d-1}] + \cdots + c_1[t] + c_0 \\
&\quad \forall [t^k] \exists \text{ representative } t^k + mh_k \\
&\quad t^d + mh_d + c_{d-1}(t^{d-1} + mh_{d-1}) + \cdots + c_1(t + mh_1) + c_0 = \\
&\quad (t^d + c_{d-1}t^{d-1} + \cdots + c_1t + c_0) + m(h_d + c_{d-1}h_{d-1} + \cdots + c_1h_1) = m + m(\cdots)
\end{aligned}$$

therefore $m(\alpha) = [0]$.

d) From 5.a every $[f]$ has unique representative $f(t) = a_0 + a_1t + \cdots + a_{d-1}t^{d-1}$.

$$f(\alpha) = a_0 + a_1\alpha + \cdots + a_{d-1}\alpha^{d-1}$$

analogous to 5.c

$$\begin{aligned}
a_{d-1}t^{d-1} + a_{d-2}t^{d-2} + \cdots + a_0 + m(a_{d-1}h_{d-1} + a_{d-2}h_{d-2} + \cdots + a_1h_1) &\iff \\
f + m(a_{d-1}h_{d-1} + a_{d-2}h_{d-2} + \cdots + a_1h_1) &\iff [f]
\end{aligned}$$

e) Copy of $\mathbb{F}_p$ is the following set: $FC = \{[f] \in K, f(t) = a_0, a_0 \in \mathbb{F}_p\}$. Prove bijection between $\mathbb{F}_p$ and FC :

•

$$\begin{aligned}
\iota: \mathbb{F}_p &\rightarrow FC, a \mapsto [a] \\
\iota(a + b) &= [a + b] = [a] + [b] = \iota(a) + \iota(b) \\
\iota(ab) &= [ab] = [a][b] = \iota(a)\iota(b)
\end{aligned}$$

Assume $a \neq b, [a] = [b]$.

$$\begin{aligned}
[a] &= [b] \iff m \mid a - b \\
\deg(a) = 0, \deg(b) = 0, \deg(a - b) = 0, \deg(m) \geq 1 &\implies m \nmid (a - b)
\end{aligned}$$

contradiction. Injection is proven.

- Surjectivity is correct by construction, since every $f(t)$ is defined as a constant polynomial where constant is from $\mathbb{F}_p$.

From 5.b $|K| = p^d, |FC| = p, d > 1 \implies |K \setminus FC| = p^d - p > 0 \implies K \not\cong \mathbb{F}_p$.